Version 2 – Tailored Report

1. Vessel Specification and Calibration

- **MARUS Coordinate Convention:** +Z forward (surge), +X starboard, +Y up.
- **Heading Convention:** Unity yaw decreases for starboard; ITTC convention is starboard = positive. The scripts and the matric-extraction notebook use the relation relPsiNautical = −(ψ- ψ0).
- **MARUS Vessels Specifications:**

Table A. MARUS boat specifications.

Length (≈ LPP along +Z) [m]	Beam (≈ width along +X) [m]	Draft (≈ height along +Y) [m]
2.952	1.468	1.361

  - Throughout this report, **L** and **LPP** have been used interchangeably.
- **Calibration & Target Speeds:** The vessel was calibrated using the standard 85% Maximum Continuous Rating (MCR) approach to determine the benchmark operational speed (V_85).

Table B. Speed calibration summary

Steady speed at 85% MCR, V_85 [m/s]	Target IMO test speed (≈ 90% of V_85), V_target [m/s]
0.923496	0.831147

2. Manoeuvre Automation

The MARUS boat model used here does not have a physical rudder. Therefore, the maneuvers are executed via a rudder-equivalent command δ realized via differential thrust (port/starboard). For the sake of traceability, two rudder signals are logged:

- Ordered δ_cmd (deg): the maneuver definition (±35° hard-over for turning circle or ±10°/±20° per zig-zag leg)
- Executed δ_exec (deg): inferred from applied thrust differential (ratio-based)

3. Speed Configurations and PID Control

To satisfy ITTC standards, the vessel must maintain a steady approach speed and shaft RPM before the rudder execute command is issued. Therefore, the Turning Circle (TC) and Zig-Zag (ZZ) implementations use a SpeedHold control mode based on a PID controller.

- **Speed PID Settings:** K_p=0.8, K_i=0.05, K_d=0.0
- **Targeting Logic:** The script holds the vessel at 0.8311 m/s for a minimum of 120 seconds.
- **Steady Window:** The manoeuvre only triggers once the Yaw Rate (r) is < 0.3°/s and the Speed Error is < 0.05 m/s for 10 consecutive seconds.

4. TurningCircleTest: stages and stop conditions

Phases:

A) Establishing the ITTC/IMO **Initial Condition**: straight run until steady yaw rate, and steady speed (SpeedHold) over a steady window.

B) **Turn**: automatically applies a hard-over command of δ_cmd = ±35° via a differential thrust proxy.

C) **Pull Out**: returns ordered δ_cmd to 0 for 10 seconds to evaluate directional stability.

D) **End Criteria:** The test terminates after a heading change of **540°** (1.5 circles), ensuring enough steady-state data for accurate System Identification (SI). Safety caps have also been integrated for the ending criteria: maxHeadingChangeDeg = 720° and maxTestTimeSeconds = 1200 s.

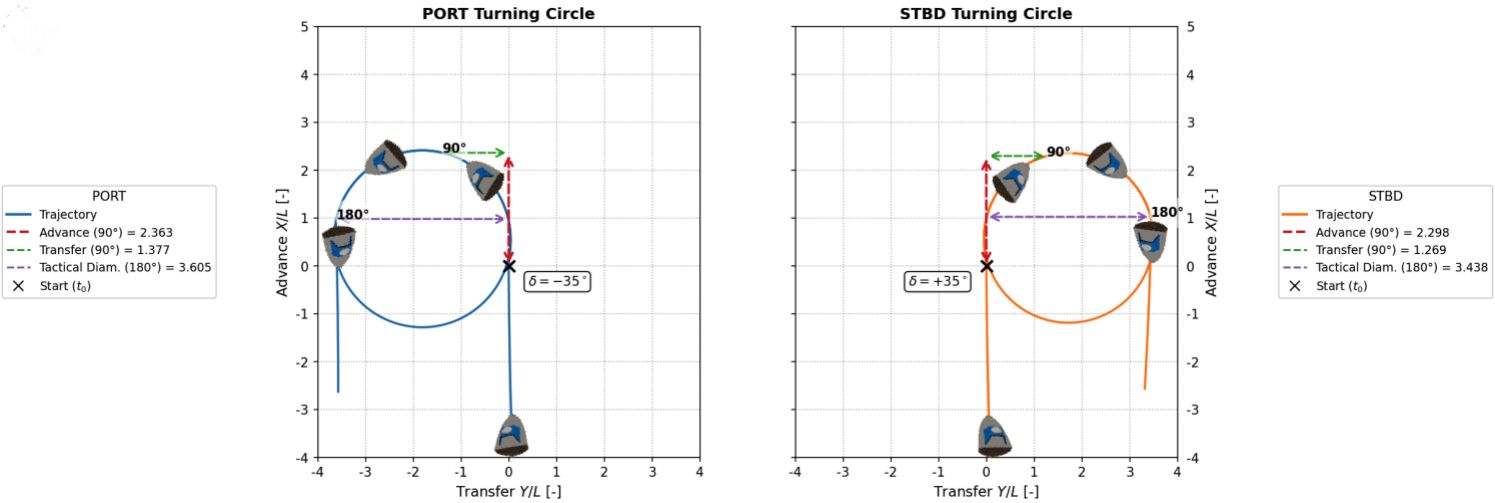


Figure 1: Turning Circle Trajectory Plots

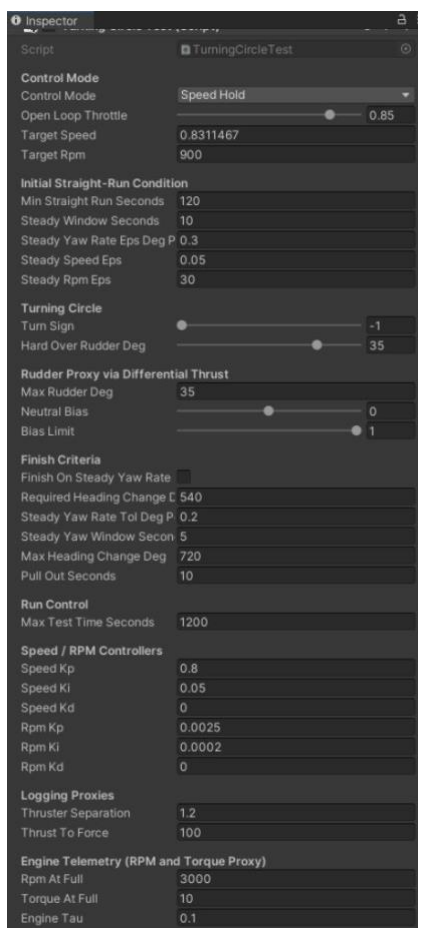


Figure 2:. The TurningCircleTest inspector configuration used for port/starboard turning circle runs.

5. Extracted TC Metrics:

- Turning circle: 1. Advance (Ad) at 90°, 2. Transfer (Tr) at 90°, 3. Tactical Diameter (Dt) at 180°, and a 4. steady diameter (D) estimate, 5. & 6. elapsed time from execute until reaching 90° and 180° heading change in seconds, and 7. maximum yaw rate in degrees per second.

Table C. Turning Circle core metrics

Run	Turn	1.Advance Ad/L	2.Transfer Tr/L	3.Tactical Diameter Dt/L	4.Steady Diameter* D/L	5. t90 (s)	6.t180 (s)	7.Max Yaw Rate (deg/s)
TC_PT.csv	PORT	2.298	1.269	3.441	3.544	14.220	28.340	6.445
TC_SB.csv	STBD	2.365	1.385	3.609	3.695	14.880	29.560	6.185

Table D. Turning Circle criteria evaluation

Run	Advance Ad/L	IMO MSC.137 Criteria Ad/L ≤ 4.5	Tactical Diameter Dt/L	IMO MSC.137 Criteria Dt/L ≤ 5.0
TC_PT.csv	2.298	PASS	3.441	PASS
TC_SB.csv	2.365	PASS	3.609	PASS

Table E. Turning Circle cross-track (XTE) diagnostics

Run	Turn	XTE* at 360° (m)	XTE at 360° /L
TC_PT.csv	PORT	0.027	0.009
TC_SB.csv	STBD	0.034	0.012

***XTE @360°: cross-track displacement at 360° heading change.**

Table F. Turning Circle velocity profiles

Run	Turn	U0* (m/s)	Froude Number*	V90 */U0 (Speed Loss)	Max Yaw Rate (deg/s)
TC_PT.csv	PORT	0.833	0.155	0.678	6.445
TC_SB.csv	STBD	0.833	0.155	0.684	6.184

*** U0: the median approach speed immediately before execution.**

*** Froude number based on LPP**

*** V90/U0: ratio of speed at 90° heading change to approach speed**

Figure 3: TC Port Velocity Profiles

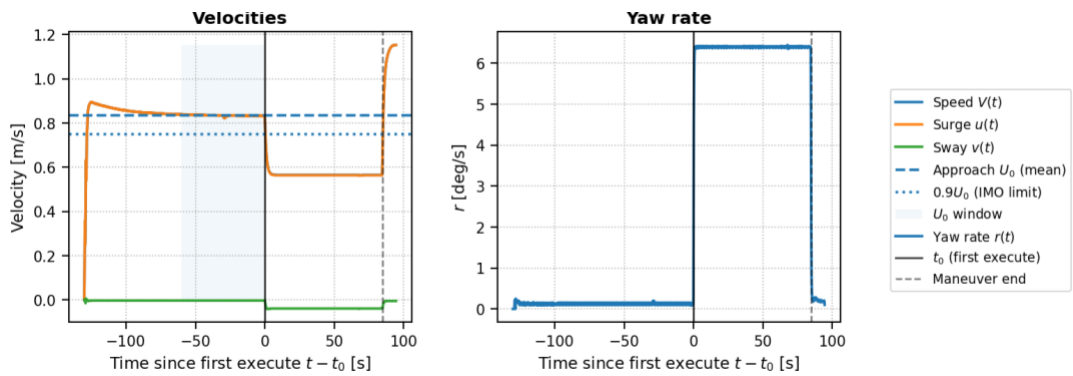
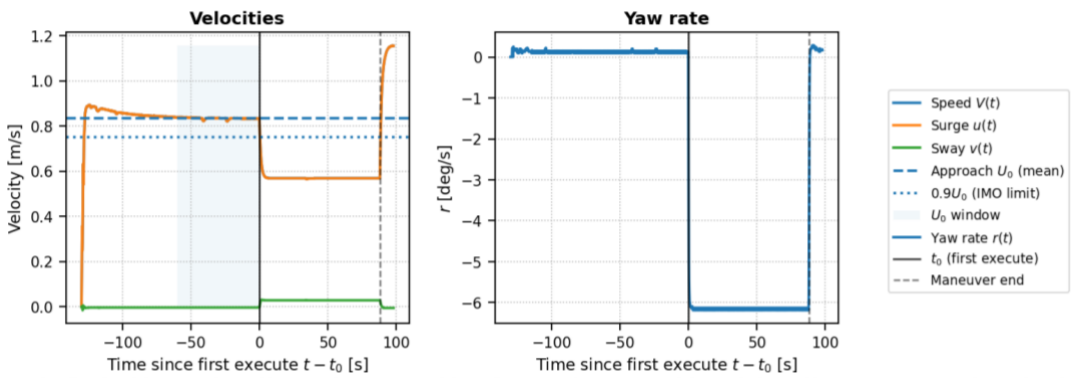


Figure 4: TC STBD Velocity Profiles



6. ZigZagManeuver: stages and stop conditions

Phases:

A) Establishing the ITTC/IMO **Initial Condition**: straight run until steady yaw rate, and steady speed (SpeedHold) over a steady window; same as TC.

B) **Zig Zag**: apply ordered δ_cmd = ±cmdAngleDeg of either 10/10 or 20/20. When the heading deflection reaches |Δψ| = cmdAngleDeg (within deadband), reverse sign.

C) **End Criteria**: stop after headingCrossingsToRun crossings (default 4 according to ITTC/IMO) or time cap of 1200 seconds.

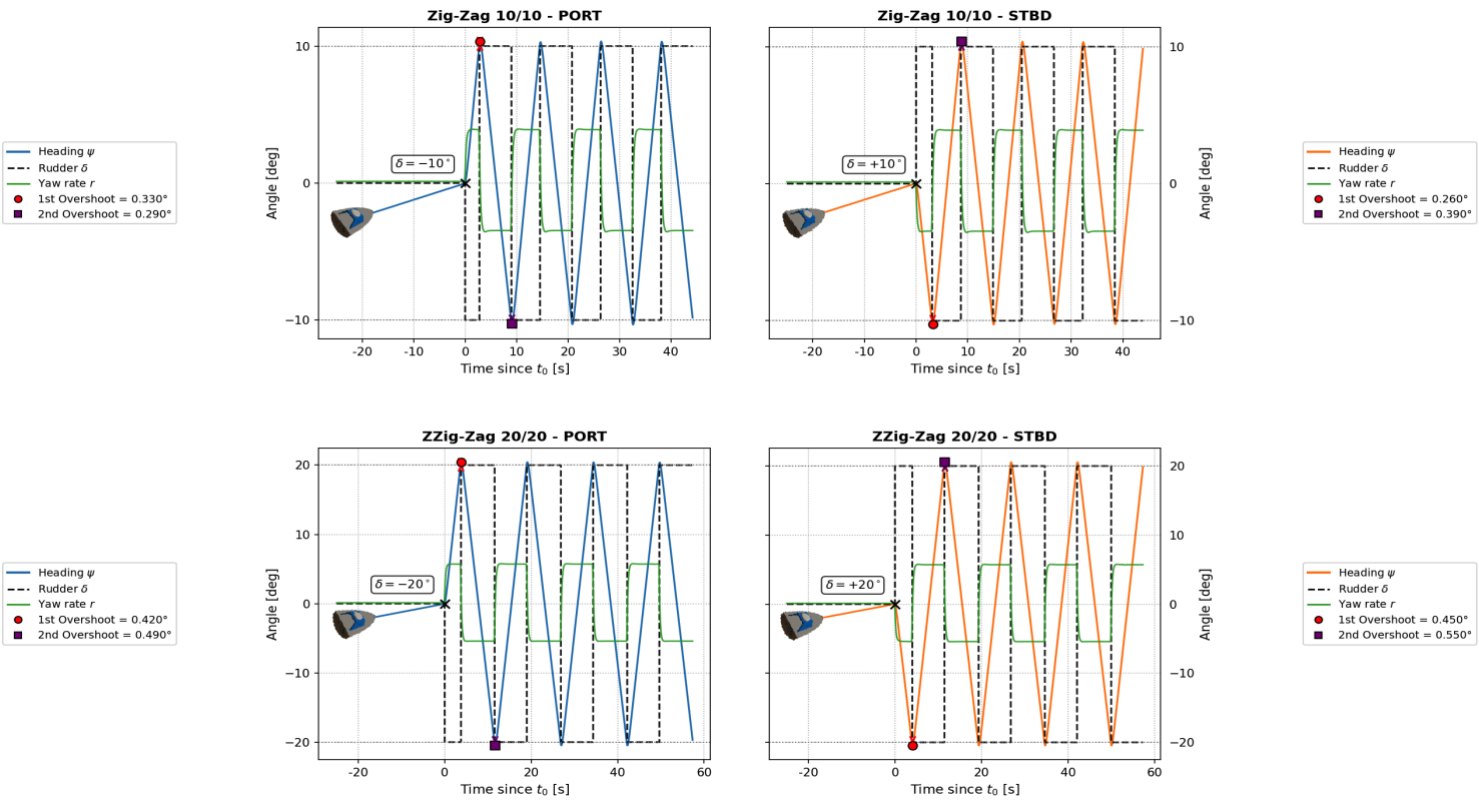


Figure 5: Zig-zag Trajectory Plots

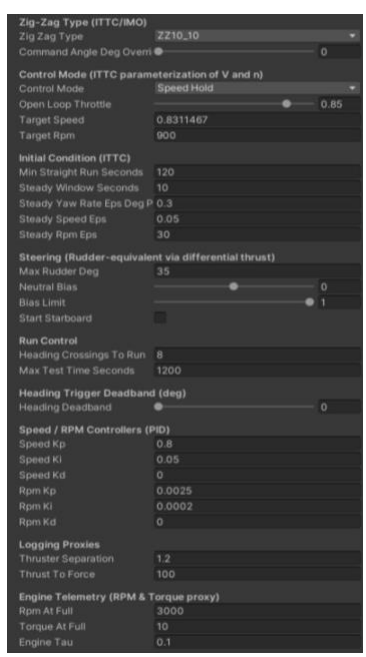


Figure 6: ZigZagManeuver inspector configuration used for ZZ10/10 and ZZ20/20 runs.

7. Extracted ZZ Metrics:

- Zig-zag: Initial turning ability (distance to 10° heading change), 1st/2nd overshoot (excess beyond cmd), and peak times.

Table G. Zig-zag core metrics

Run	Type	First-leg direction	Initial turning distance to 10° heading change/L	1st overshoot (deg)	2nd overshoot (deg)	Max yaw rate (deg/s)	Time to 1st peak heading (s)	Time to 2nd peak heading (s)
ZZ10_PT.csv	10/10	PORT	0.768	0.330	0.290	3.962	2.880	9.100
ZZ10_SB.csv	10/10	STBD	0.855	0.260	0.390	3.962	3.220	8.780
ZZ20_PT.csv	20/20	PORT	0.494	0.420	0.490	5.777	3.780	11.660
ZZ20_SB.csv	20/20	STBD	0.517	0.450	0.550	5.777	4.100	11.560

Table H. Zig-zag criteria evaluation

Run	Type	Initial turning distance to 10° heading change/L	MSC.137 Criteria ≤ 2.5	1st overshoot (deg)	Limit*	PASS	2nd overshoot (deg)	Limit*	PASS
ZZ10_PT.csv	10/10	0.768	PASS	0.330	10.000	PASS	0.290	25.000	PASS
ZZ10_SB.csv	10/10	0.855	PASS	0.260	10.000	PASS	0.390	25.000	PASS
ZZ20_PT.csv	20/20	0.494	PASS	0.420	25.000	PASS	0.490	40.000	PASS
ZZ20_SB.csv	20/20	0.517	PASS	0.450	25.000	PASS	0.550	40.000	PASS

***IMO (MSC.137) criteria used for evaluation:**
• 10°/10°: 1st overshoot ≤ 10°, 2nd overshoot ≤ 25°
• 20°/20°: 1st overshoot ≤ 25°. 2nd overshoot is reported for completeness ≤ 40°.

Table I. Zig-zag XTE

Rn	Type	Max XTE* (m)	Max XTE /L
ZZ10_PT.csv	10/10	0.483	0.163
ZZ10_SB.csv	10/10	0.616	0.209
ZZ20_PT.csv	20/20	0.973	0.330
ZZ20_SB.csv	20/20	1.208	0.409

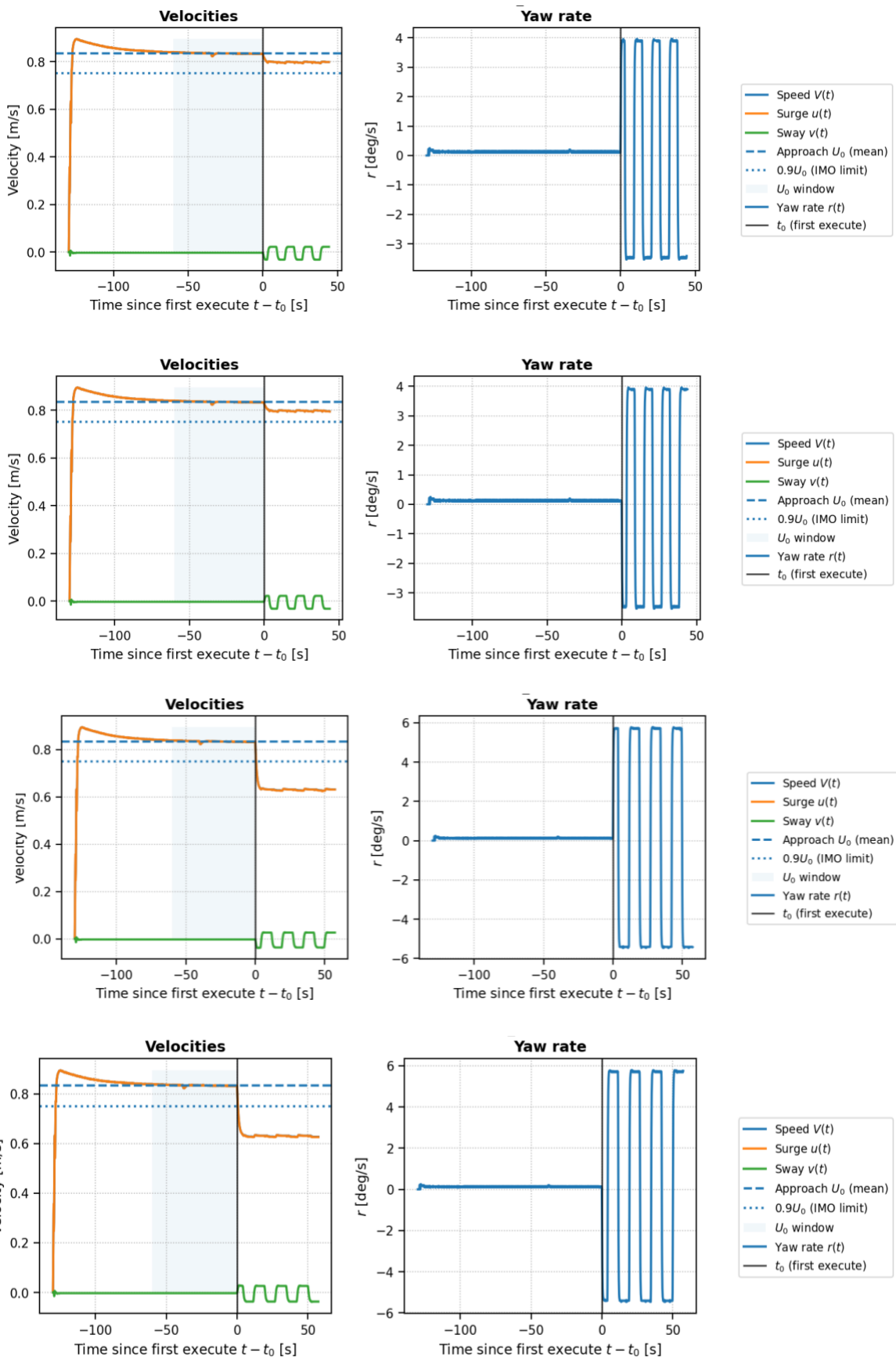
***Max XTE (m): Maximum absolute cross-track displacement during the zig-zag run, computed by projecting the vessel's position onto the initial starboard-positive perpendicular axis defined at manoeuvre start.**

Table J. Zig-zag velocity profiles

Run	Type	U0* (m/s)	Froude Number*	Normalized speed at 1st peak	Normalized speed at 2nd peak	Max Yaw Rate (deg/s)
ZZ10_PT.csv	10/10	0.833	0.155	0.957	0.957	3.962
ZZ10_SB.csv	10/10	0.833	0.155	0.959	0.954	3.962
ZZ20_PT.csv	20/20	0.833	0.155	0.761	0.758	5.777
ZZ20_SB.csv	20/20	0.833	0.155	0.763	0.752	5.777

*** U0: the median approach speed immediately before execution.**

Figure 7: ZZ Velocity Profiles in the order: 1. ZZ10/10 Port 2. ZZ10/10 STBD 3. ZZ20/20 Port 4.ZZ20/20 STBD



References

- International Towing Tank Conference. (n.d.). *Recommended procedures and guidelines: Manoeuvring – Turning circle and zig-zag tests (Procedure No. 7.5-02-06-03)*
- International Maritime Organization. (2002). *Resolution MSC.137(76): Standards for ship manoeuvrability*